CIS4910 Literature Review

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Introduction

Following the development of a segmentation model to identify the navel in previous work, the broader pipeline required to track the motion of an insulin pen was considered, with the ultimate goal of localizing injection sites on the abdomen. While it may be feasible to combine this segmentation model with geometric methods to estimate the pen's pose over time, such approaches typically require precise calibration of cameras and inertial sensors, as is common in classical geometry-based visual-inertial odometry (VIO).

To address this limitation, the possibility of using an end-to-end trainable neural network for VIO was explored, thereby reducing or eliminating the need for manual calibration. Transformers, through their attention mechanisms, have demonstrated strong adaptability across modalities beyond text. For instance, the Vision Transformer (ViT) architecture [1] achieves state-of-the-art performance in image classification, surpassing convolutional neural networks that have historically dominated the field. Motivated by this, the basic transformer architecture was investigated before moving on to prior work [2] that employs a transformer encoder for deep VIO.

1 Background

1.1 Transformers

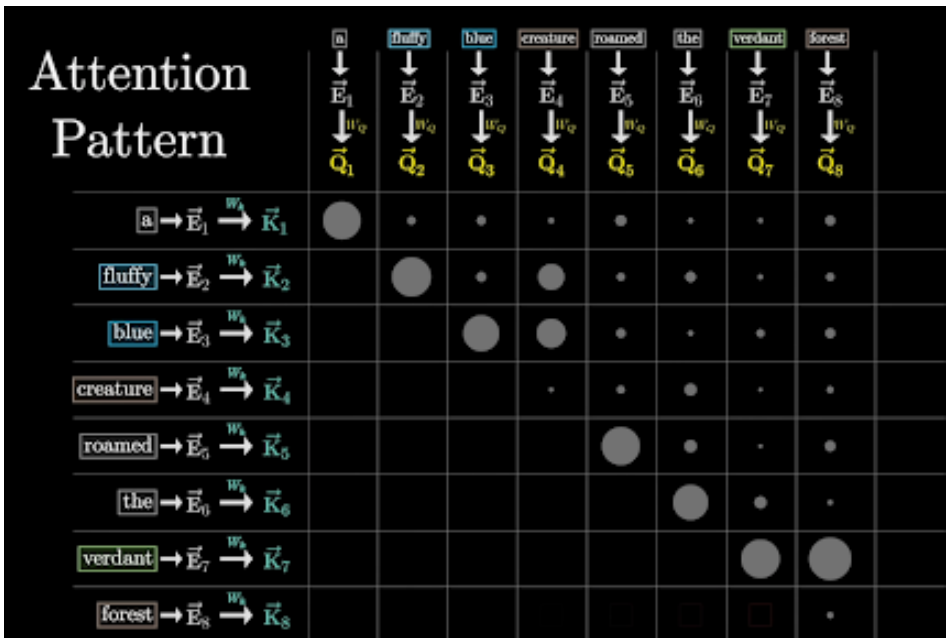

Figure 1: Visualization of self-attention patterns between tokens, illustrating how each token attends to others through query–key interactions. Adapted from [3].

Although the overall transformer architecture is not conceptually complex, its breadth and scale make it challenging to develop an intuitive understanding of the internal computations performed by the model. In addition to the primary literature, educational resources such as the 3Blue1Brown video series on large language models were consulted to supplement this understanding. These resources were particularly helpful in developing an intuitive understanding of the attention mechanism and how the query, key, and value matrices interact to allow information to flow between tokens. Furthermore, the visualizations aided in understanding masked attention, multi-head attention blocks, and the purpose of the feed-forward network layers interleaved between attention blocks. This foundational review supported the interpretation of architectures discussed in this work, particularly the transformer encoder used in the VIFT architecture.

1.2 Transformer-based Visual-Inertial Odometry (VIFT)

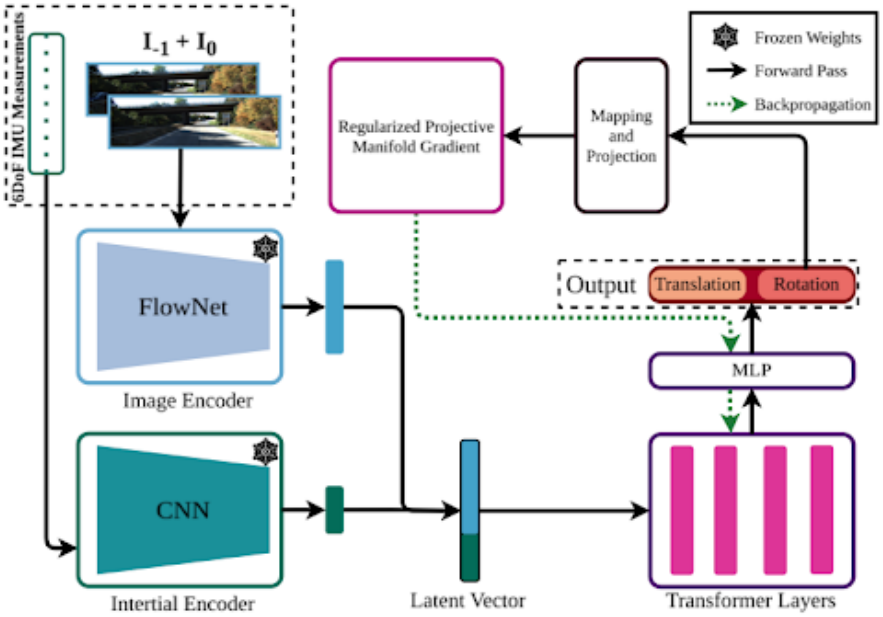

Figure 2: Overview of the VIFT architecture, adapted from [2].

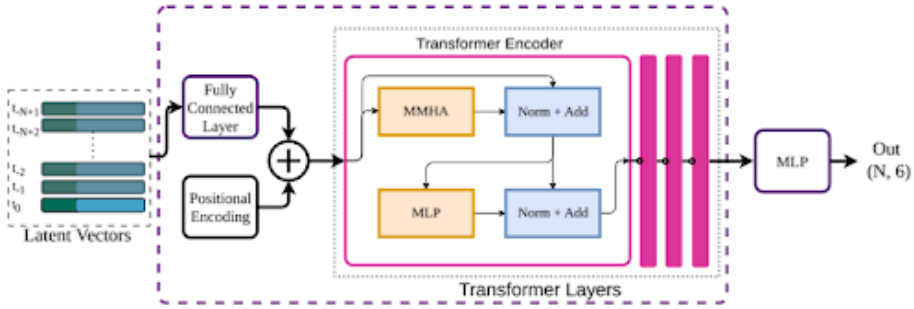

Figure 3: Transformer encoder architecture used in the VIFT model, adapted from [2].

The VIFT architecture [2] employs a transformer encoder to refine pose estimates via temporal attention over visual-inertial feature representations, as illustrated in Figures 2 and 3. Although the method produces strong results, the manner in which visual and inertial data are incorporated into the network appears relatively simplistic. Two consecutive visual frames are processed using a pre-trained CNN, while the intervening IMU data are passed through a 1D CNN; the resulting feature representations are then concatenated. This design may limit the model's ability to capture structured cross-modal interactions, as concatenation alone does not explicitly model relationships between visual and inertial features.

However, the authors follow the recommendations of [1] by leveraging convolutional inductive bias rather than directly processing raw inputs. This strategy has been shown to improve the

performance of transformers on smaller datasets, where available data may be insufficient for such models to learn relevant patterns directly from raw inputs. These observations motivate the question of whether performance could be improved through a more principled approach to multimodal fusion within the transformer encoder. Accordingly, this work explores alternative fusion mechanisms that may enable more robust and structured integration of visual and inertial information.

2 Proposed Directions

This section explores several potential directions for improving the VIFT architecture, including alternatives to the existing concatenation-based fusion method, strategies for improving representation learning through contrastive pretraining, and methods for enhancing training stability and efficiency.

2.1 Joint Attention for Multimodal Fusion

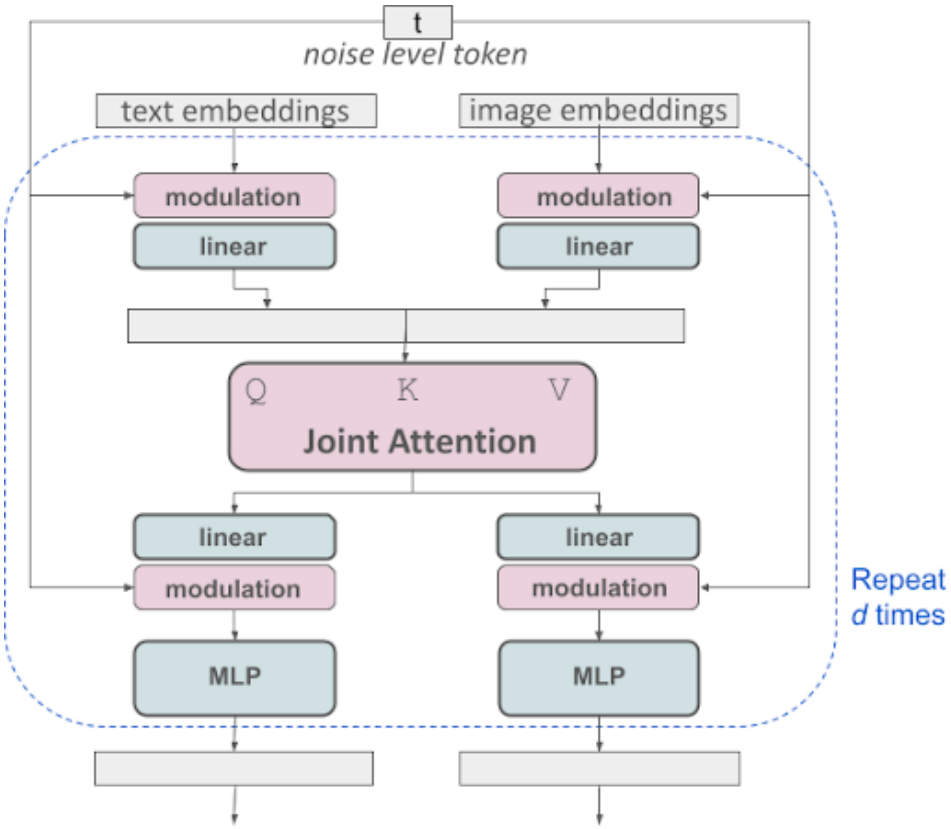

Figure 4: Joint attention block in the MM-DiT architecture. Adapted from [4].

The Multimodal Diffusion Transformer (MM-DiT) [4] fuses visual and textual information to generate images from textual prompts. As shown in Figure 4, this architecture introduces a joint attention mechanism that allows each modality to operate within its own representation space while still incorporating information from the other. This design suggests a natural

extension to deep VIO, where each modality first attends within its own domain before engaging in cross-modal interactions.

This perspective highlights a potential limitation in the VIFT model [2], namely an imbalance between modality representations: the visual latent has dimensionality 512, while the IMU latent has dimensionality 256. As a result, a larger proportion of the representational capacity may be associated with the visual modality, which could bias the model toward visual features due to its greater latent dimensionality and its influence on how information is distributed across shared attention projections. To complement the primary literature, supplementary explanatory resources were consulted, including a video by Julia Turc [5], which provided useful intuition for the joint attention mechanism used in MM-DiT.

2.2 Cross-Attention for Visual-Inertial Fusion

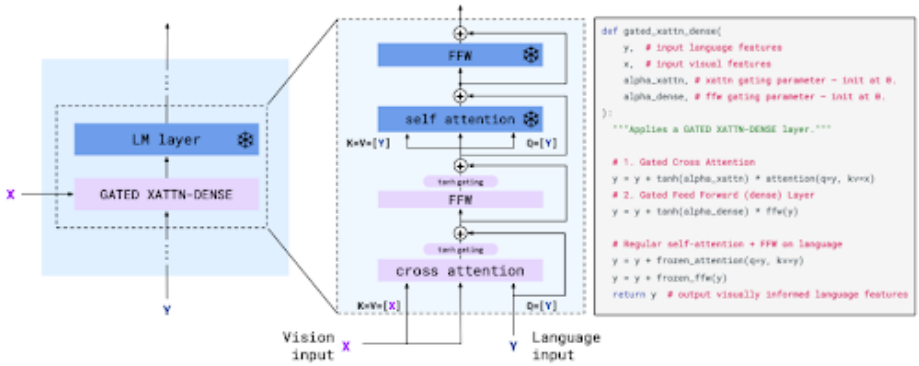

Figure 5: Gated cross-attention mechanism from the Flamingo architecture, adapted from [6].

Another approach is to employ cross-attention to enable interaction between modalities. Cross-attention can be applied either symmetrically or asymmetrically. In the symmetric case, each modality attends to the other, allowing visual features to query IMU representations and vice versa. In the asymmetric case, the IMU—being more directly related to motion measurements across frames—can query visual features to correct for drift, a common issue in inertial sensing. In both settings, cross-attention enables more flexible modeling of cross-modal relationships compared to simple concatenation. However, the asymmetric formulation may be particularly suitable, as IMU signals provide temporally consistent motion information that can complement visual observations.

As shown in Figure 5, this approach is supported by prior work such as Flamingo, which introduces a multimodal visual-language model that uses cross-attention from language tokens to visual features to condition text generation. A similar mechanism can be applied in the VIO setting, where IMU representations query visual features to improve robustness and mitigate drift.

2.3 Contrastive Pretraining for Modality Alignment

As previously described, VIFT [2] concatenates visual and inertial features, projects them through a linear layer, and feeds the result into a transformer encoder. One potential

extension is to pre-train separate encoders using a contrastive objective to align visual and IMU representations in a shared latent space prior to temporal modeling, similar to the CLIP model [7]. This may enable more effective cross-modal interactions when the representations are subsequently processed by the transformer.

Although this approach increases the overall model complexity, the encoders may learn to align the two modalities in latent space, allowing attention to operate on representations that share similar latent structure. Additionally, pre-training the encoders may simplify optimization compared to training the full model end-to-end to simultaneously learn both modality alignment and temporal attention. However, this strategy may also introduce bias by constraining the representations to align across modalities, rather than allowing the model to learn representations that are optimally suited for pose estimation.

2.4 Alternative Optimizers for Transformer Training

Recent work [8] suggests that the Muon optimizer can improve training efficiency and convergence in transformer-based models. This improvement is attributed to modifications in how updates are applied to weight matrices during optimization, which may lead to more stable training dynamics. It may therefore be worthwhile to evaluate whether replacing the AdamW optimizer used in VIFT with Muon leads to improved training stability or faster convergence in the VIO setting. However, the effectiveness of such optimizers may depend on factors such as dataset size and model scale, and their impact on multimodal VIO architectures remains largely unexplored.

3 Conclusion

In this work, several potential directions for improving transformer-based visual-inertial odometry have been explored, including joint attention, cross-attention, contrastive representation alignment, and alternative optimization strategies. These approaches aim to provide alternatives to the existing concatenation-based fusion method used in VIFT while also improving training stability and efficiency. Future work will involve evaluating these methods to determine which approach provides the most effective integration of visual and inertial information for accurate pose estimation.

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References

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